Product Sum

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The problem

- A "special" array holds ints or nested special arrays; each nested array's sum is multiplied by its depth (outermost = 1).

Intuition

- A nested array's sum is multiplied by its depth, which is naturally recursive.

Approach

- 1. Sum the elements at this level
- 2. For a nested list, recurse with depth plus one
- 3. Multiply the level sum by the depth
- 4. Return it

Java solution

```java
public class ProductSum {
    public static int productSum(List<Object> array) { return productSum(array, 1); }

    @SuppressWarnings("unchecked")
    private static int productSum(List<Object> array, int depth) {
        int sum = 0;
        for (Object o : array) sum += (o instanceof List) ? productSum((List<Object>) o, depth + 1) : (Integer) o;
        return sum * depth;
    }
}
```

Complexity

- Time: O(n) total elements
- Space: O(d) depth

Recap

- A nested array's sum is multiplied by its depth, which is naturally recursive.
- Complexity: O(n) total elements